Bounds, Signs, and the Variable of Integration: Find the Error

- Most problems below show a student's work. The antiderivatives are correct — the mistakes are in the *limits*, the *sign*, or the *variable*.
- When a part asks you to name or describe an error, write a sentence.
- Give exact values (fractions, radicals) unless a decimal is requested.

1. Evaluate $\int_0^3 (2x - 4)\,dx$. Your answer is a signed value, so keep the sign.

Answer: ________________

2. Asked for $\int_5^2 2x\,dx$, a student writes "$[x^2]$ *from* 2 *to* 5 *is* $25 - 4 = 21$."

(a) Name the error.

Answer: ________________

(b) Give the corrected value.

Answer: ________________

3. Asked for $\dfrac{d}{dx}\displaystyle\int_1^x (3t^2+1)\,dt$, a student answers "*$3t^2+1$, so at $x=2$ the value is 4.*"

(a) Name the error.

Answer: ____________________

(b) Give the corrected value at $x = 2$.

Answer: ____________________

4. Asked for $\displaystyle\int_0^2 x\left(x^2+1\right)^3 dx$, a student substitutes $u = x^2+1$, $du = 2x\,dx$, and writes

$$\frac{1}{2}\int_0^2 u^3\,du = \frac{1}{8}\left[u^4\right]_0^2 = 2.$$

(a) Name the error.

Answer: ____________________

(b) Give the corrected value.

Answer: ____________________

5. A particle moves with velocity $v(t) = t^2 - 4$ metres per second for $0 \leq t \leq 3$. A student reports the *total distance travelled* as $\displaystyle\int_0^3 (t^2 - 4)\, dt = -3$ metres.

(a) Describe the error.

Answer: ________________

(b) Find $\displaystyle\int_0^2 (4 - t^2)\, dt$.

Answer: ________________

(c) Find $\displaystyle\int_2^3 (t^2 - 4)\, dt$.

Answer: ________________

(d) Give the total distance travelled.

Answer: ________________

6. Asked for $\int_0^{\sqrt{\pi/2}} x\cos\left(x^2\right)dx$, a student substitutes $u = x^2$, replaces $x\,dx$ by du, and reports the value 1.

(a) Name the error.

Answer: ______________

(b) Give the corrected value.

Answer: ______________

7. Asked for the *area* between $y = x^3$ and the x-axis on $[-2, 2]$, a student computes $\int_{-2}^{2} x^3\,dx = 0$ and concludes the area is 0.

(a) Describe the error.

Answer: ______________

(b) Give the corrected area.

Answer: ______________

8. Asked for $\int_1^4 \frac{1}{\sqrt{x}}\,dx$, a student substitutes $u = \sqrt{x}$, so $du = \frac{1}{2\sqrt{x}}\,dx$, and writes $\int_1^4 2\,du = 2(4-1) = 6$.

(a) Name the error.

Answer: ____________

(b) Give the corrected value.

Answer: ____________

(c) Describe the habit that prevents this error from happening again.

Answer: ____________